

## Contribution Journal – Module 03 Machine Learning Lab

For this assignment, I worked alone on the entire Module 03 machine learning lab. I normally prefer working independently because it allows me to move at my own pace, focus without distractions, and make sure every part of the assignment is completed correctly. I also had difficulty finding a group to work with, so completing the lab on my own was the most practical option. Working solo meant that I was responsible for every step of the project, from understanding the instructions to running the code and writing the reflections.

My main contribution to the project was completing all parts of the lab myself. I handled the data loading, exploration, and visualization. I examined the Wine dataset, looked at correlations, and reviewed how the features were distributed. I also completed the machine learning workflow by splitting the data, training the models, and evaluating their performance. I trained both Logistic Regression and Decision Tree models and compared their results. I interpreted the confusion matrix and accuracy scores and experimented with different features to see how they affected the model's performance. All of the coding, analysis, and reflection writing was done by me.

I also contributed by managing the workflow and keeping the assignment organized. I made sure each part of the lab was completed in the correct order and that the results made sense. I wrote the Reflection and Analysis section in the notebook and answered the reflection questions based on what I learned. I took responsibility for understanding the machine learning concepts, reviewing the output, and making sure the final results were accurate. Working alone required me to stay focused and solve problems without relying on teammates, which helped me build confidence in my ability to complete technical tasks independently.

Throughout the project, I used several skills. I used Python and scikit-learn to run the models and analyze the data. I used data exploration skills to understand the dataset and identify important features. I used problem-solving skills to interpret the confusion matrix and figure out why certain classes were misclassified. I also used writing and communication skills to explain my results clearly in the reflection sections. Managing the entire project alone helped me strengthen these skills and become more comfortable with the machine learning workflow.

Working alone also came with challenges. The most difficult part was interpreting the confusion matrix and understanding why class\_2 had more errors. Without teammates to discuss the results with, I had to rely on my own reasoning and go back through the data to understand what was happening. Another challenge was making sure I understood each step of the workflow, especially when switching between data exploration, model training, and evaluation. Even though it was challenging, completing the lab independently helped me learn the material more deeply.

If I were working in a team, I would contribute by helping organize the workflow, explaining concepts to others, and assisting with coding and analysis. I would also help write reflections and make sure the results were clearly communicated. Even though I worked alone for this assignment, I know I could contribute effectively in a group setting by staying organized, communicating clearly, and helping solve problems.

Overall, completing this lab on my own helped me understand the machine learning process from start to finish. I contributed by completing every part of the assignment myself, managing the workflow, solving challenges, and writing clear reflections. This experience helped me build confidence in my abilities and prepared me for future assignments in the course.